Combining Measurement and Latent Variable Models

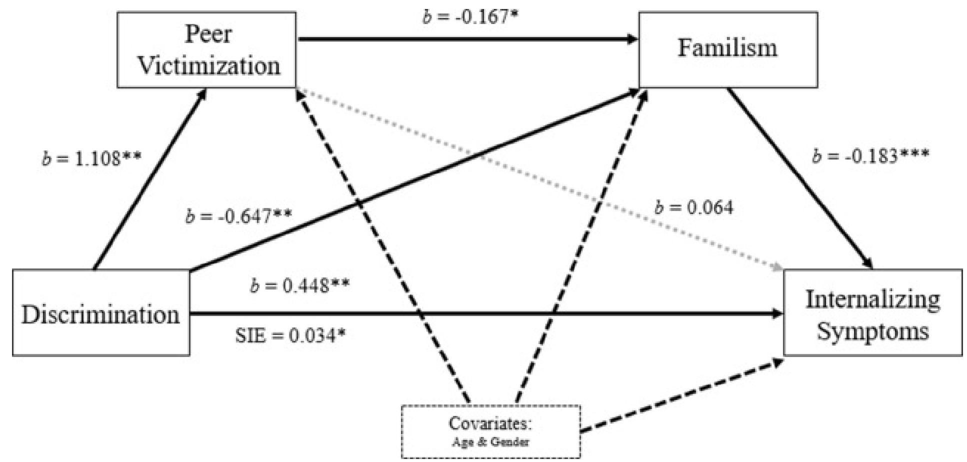
Agenda

- Specification
- Identification
- Estimation
- Evaluation
- Modification
- Interpretation
- Model Comparison

Understanding Internalizing Symptoms for LatinX Youth

Delgadillo, D., Ramos, G., Montoya, A. K., Thamrin, H., Rapp, A., Escovar, E., & Chavira, D. (2021). Discrimination and Internalizing Disorders in Rural Latinx Adolescents: An Ecological Model of Etiology. *Children and Youth Services Review*, *130*, 106250.

639 LatinX students age 14 – 17 completed surveys on multiple measures



Age (in years)

Gender (Male/Female)

Perceived Discrimination: 9 Items SAFE-C on a scale of 1-4

Peer Victimization: 4 items Illinois Bullying Scale (IBS) on a scale of 1-4

Familism: 7 items Familism Scale (FS) on a scale of 1-5

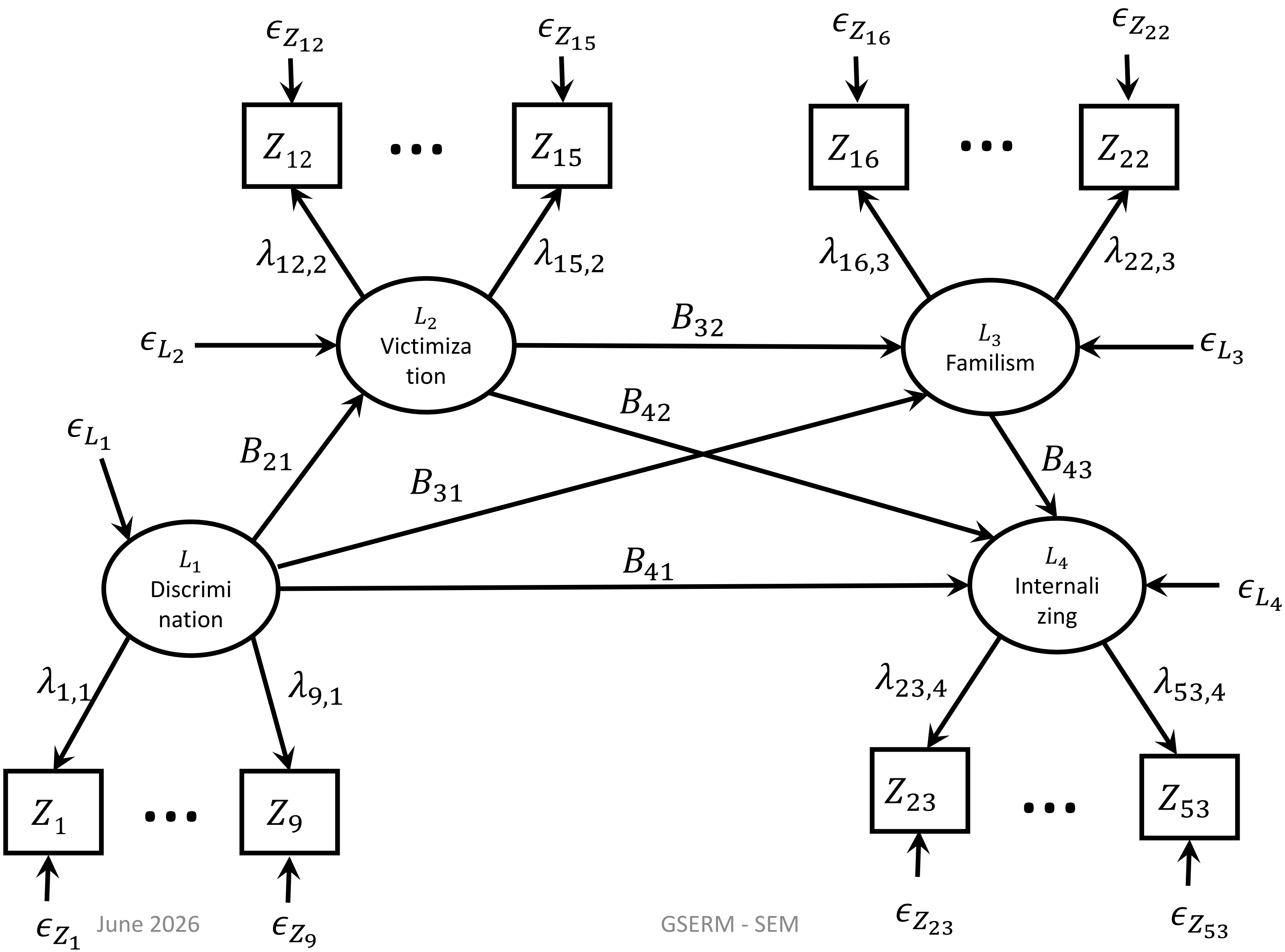
Internalizing Symptoms: 31 items Youth Self Report (YSR) on a scale of 0-2.

Research Questions

- Does a model where discrimination affects peer victimization, familism, and internalizing; peer victimization affects familism and internalizing; and familism affects internalizing **fit the data well**?
- What is the **indirect effect** of discrimination on internalizing symptoms through peer victimization and familism in serial?
- Does a model which only allows for the serial indirect effect of discrimination on familism, and no other indirect (or direct) effects, fit better/well?

Specification

Specification



Incorporating Observed Covariates in Notation

Age and Gender are observed covariates, which we assume have no measurement error

Our notation does not allow for observed variables to affect latent variables directly

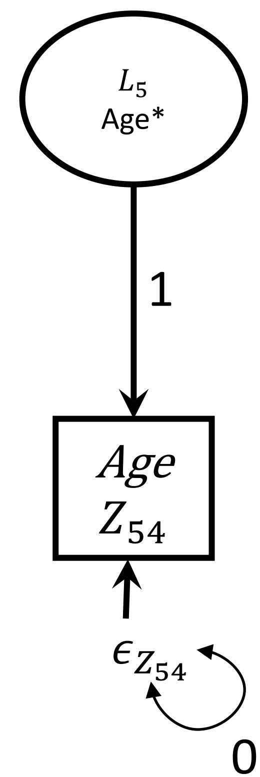
These can be specified in our notation (a little clunkily)

We can assume there is a "latent variable" age L_5

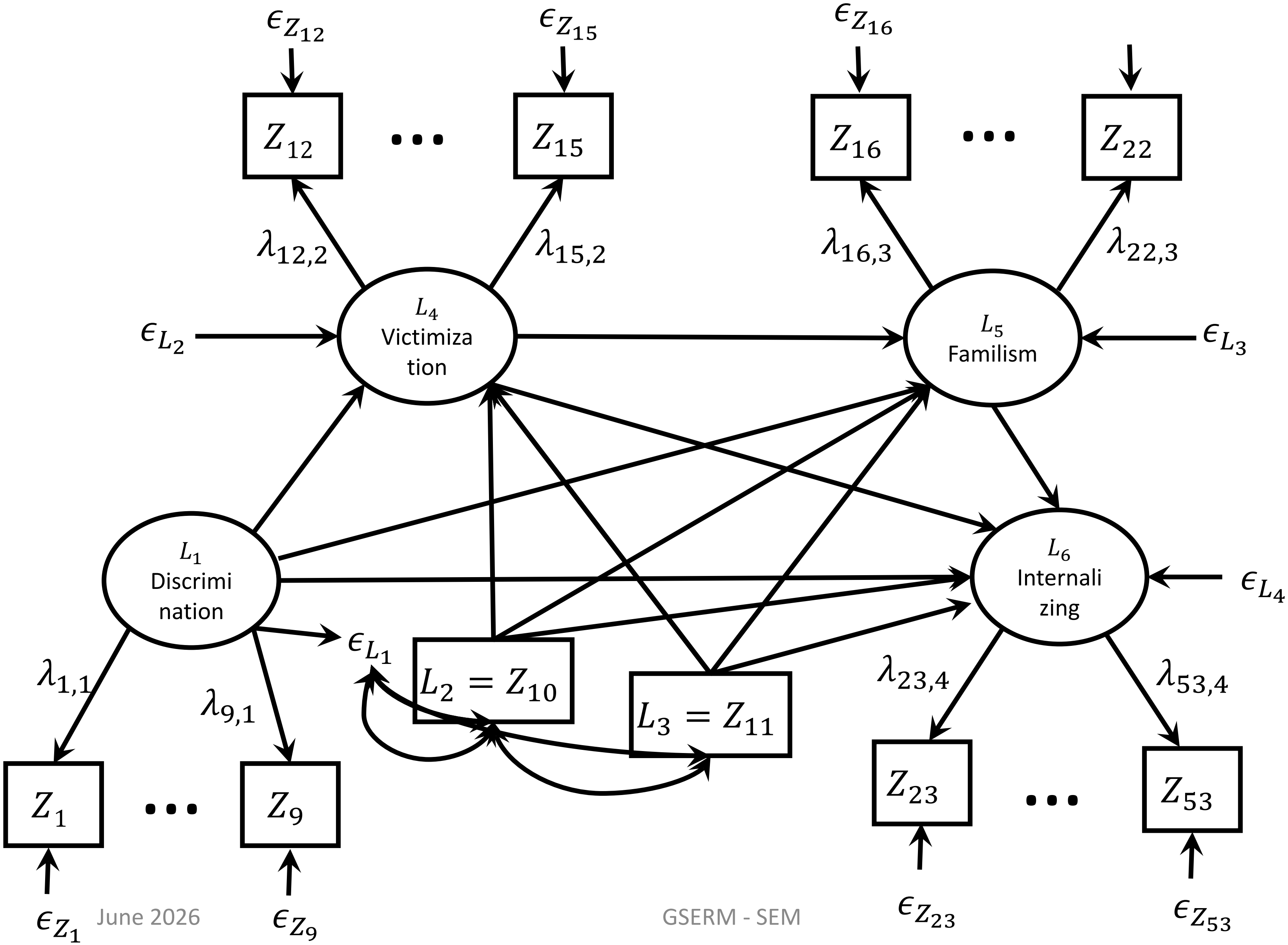
This latent variable affects the observed variable age

The observed variable age is a perfect indicator:

- Loading = 1
- Variance of deviance = 0



Specification



Measurement Equations

The measurement models are all relatively simple:

Discrimination

$$Z_1 = \alpha_{Z_1} + \lambda_{1,1} L_1 + \epsilon_{Z_1}$$

$$\vdots$$

$$Z_9 = \alpha_{Z_9} + \lambda_{9,1} L_1 + \epsilon_{Z_9}$$

Peer Victimization

$$Z_{12} = \alpha_{Z_{12}} + \lambda_{12,2} L_2 + \epsilon_{Z_{12}}$$

$$\vdots$$

$$Z_{15} = \alpha_{Z_{15}} + \lambda_{15,2} L_2 + \epsilon_{Z_{15}}$$

Familism

$$Z_{16} = \alpha_{Z_{16}} + \lambda_{16,3} L_3 + \epsilon_{Z_{16}}$$

$$\vdots$$

$$Z_{22} = \alpha_{Z_{22}} + \lambda_{22,3} L_3 + \epsilon_{Z_{22}}$$

Internalizing Symptoms

$$Z_{23} = \alpha_{Z_{23}} + \lambda_{23,4} L_4 + \epsilon_{Z_{23}}$$

$$\vdots$$

$$Z_{53} = \alpha_{Z_{53}} + \lambda_{53,4} L_4 + \epsilon_{Z_{53}}$$

Latent Variable Equations

Without Covariates:

$$L_{1i} = \alpha_{L_1} + \epsilon_{L_{1i}}$$

$$L_{2i} = \alpha_{L_2} + B_{21}L_{1i} + \epsilon_{L_{2i}}$$

$$L_{3i} = \alpha_{L_3} + B_{31}L_{1i} + B_{32}L_{2i} + \epsilon_{L_{3i}}$$

$$L_{4i} = \alpha_{L_4} + B_{41}L_{1i} + B_{42}L_{2i} + B_{43}L_{3i} + \epsilon_{L_{4i}}$$

With Covariates:

$$L_{1i} = \alpha_{L_1} + \epsilon_{L_{1i}}$$

$$L_{4i} = \alpha_{L_4} + B_{41}L_{1i} + B_{42}L_{2i} + B_{43}L_{3i} + \epsilon_{L_{4i}}$$

$$L_{5i} = \alpha_{L_5} + B_{51}L_{1i} + B_{52}L_{2i} + B_{53}L_{3i} + B_{54}L_{4i} + \epsilon_{L_{5i}}$$

$$L_{6i} = \alpha_{L_6} + B_{61}L_{1i} + B_{62}L_{2i} + B_{63}L_{3i} + B_{64}L_{4i} + B_{65}L_{5i} + \epsilon_{L_{6i}}$$

Factor Loading Matrix

$$\begin{array}{cccccc} L_1 & L_2 & L_3 & L_4 & L_5 & L_6 \end{array}$$

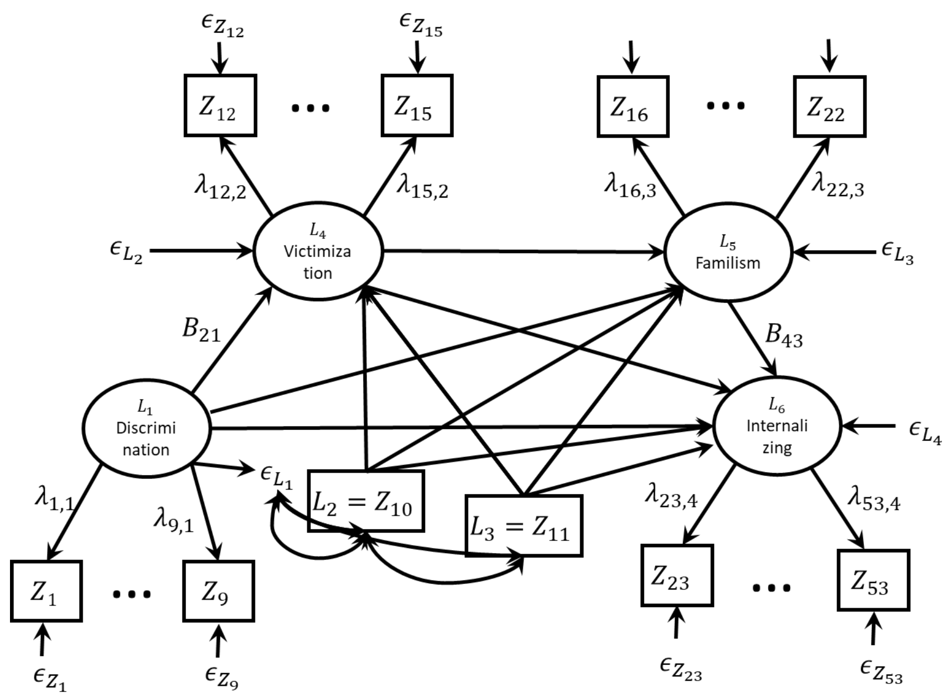
$\boldsymbol{\Lambda}_x$

$$\boldsymbol{\Lambda}_{43\times 6} = \begin{bmatrix} \\ \\ \\ \end{bmatrix}$$

$\boldsymbol{\Lambda}_y$

Latent Variable Relationship Matrices

$$\mathbf{B}_{6\times 6} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ B_{41} & B_{42} & B_{43} & 0 & 0 & 0 \\ B_{51} & B_{52} & B_{53} & B_{54} & 0 & 0 \\ B_{61} & B_{62} & B_{63} & B_{64} & B_{65} & 0 \end{bmatrix}$$



Covariance Matrices

$$\boldsymbol{\Sigma}_{\boldsymbol{\epsilon}_{\boldsymbol{Z}}} = \begin{bmatrix} \sigma^2_{\epsilon_{Z_1}} & & \\ \vdots & \ddots & \\ 0 & \cdots & \sigma^2_{\epsilon_{Z_{52}}} \end{bmatrix}$$

Diagonal elements represent variances of each deviance. There are no correlated deviances in the specification, so all off diagonal elements are 0

$$\boldsymbol{\Sigma}_{\boldsymbol{\epsilon}_{L}} = \begin{bmatrix} \boldsymbol{\Sigma}_{\boldsymbol{\epsilon}_{L_X}} & \\ \mathbf{0} & \boldsymbol{\Sigma}_{\boldsymbol{\epsilon}_{L_Y}} \end{bmatrix}$$

All exogenous variables are allowed to covary and have an estimated variance $\boldsymbol{\Sigma}_{\boldsymbol{\epsilon}_{L_X}}$

Deviances for endogenous variables each have a variance (on the diagonal) and no covariances with each other (all off diagonals are zero) $\boldsymbol{\Sigma}_{\boldsymbol{\epsilon}_{L_Y}}$

Code in lavaan

Specifying Each Factor

```
factormodel <- "

Disc =~ safe1 + safe2 + safe8 + safe10 + safe13 + safe15 + safe16 +
safe20 + safe21

Vict =~ bully4+bully5+bully6+bully7

Famil =~FS1+FS2+FS3+FS4+FS5+FS6+FS7

Internal =~
ysr_prob_14+ysr_prob_29+ysr_prob_30+ysr_prob_31+ysr_prob_32+ysr_pro
b_33+ysr_prob_35+ysr_prob_45+ysr_prob_50+ysr_prob_52+ysr_prob_71+ys
r_prob_91+ysr_prob_112+ysr_prob_5+ysr_prob_42+ysr_prob_65+ysr_prob_
69+ysr_prob_75+ysr_prob_102+ysr_prob_103+ysr_prob_111+ysr_prob_47+y
sr_prob_51+ysr_prob_54+ysr_prob_56a+ysr_prob_56b+ysr_prob_56c+ysr_p
rob_56d+ysr_prob_56e+ysr_prob_56f+ysr_prob_56g+ysr_prob_56h

"
```

Latent Variable Regressions

```
latentvariables <- "
Vict ~ Disc+Age+Gender
Famil ~ Disc+Vict+Age+Gender
Internal ~ Disc+Vict+Famil+Age+Gender
"
```

Identification

Identification for Combined Models

If an unknown parameter in $\boldsymbol{\theta}$ can be written as a function of one or more elements of $\boldsymbol{\Sigma}$, that parameter is identified. If all unknown parameters in $\boldsymbol{\theta}$ are identified, then the model is identified.

Rules for identification:

- N_θ-Rule
  - We're already familiar with this one, won't review again
- Two-Step Rule
- MIMIC Rule

Local fit evaluation from CFA unit still applies

Two-Step Rule

Sufficient but not necessary condition for identification

Step 1. Treat the model as a confirmatory factor model, is it identified? If yes, move on to Step 2.

Can apply all CFA rules here:

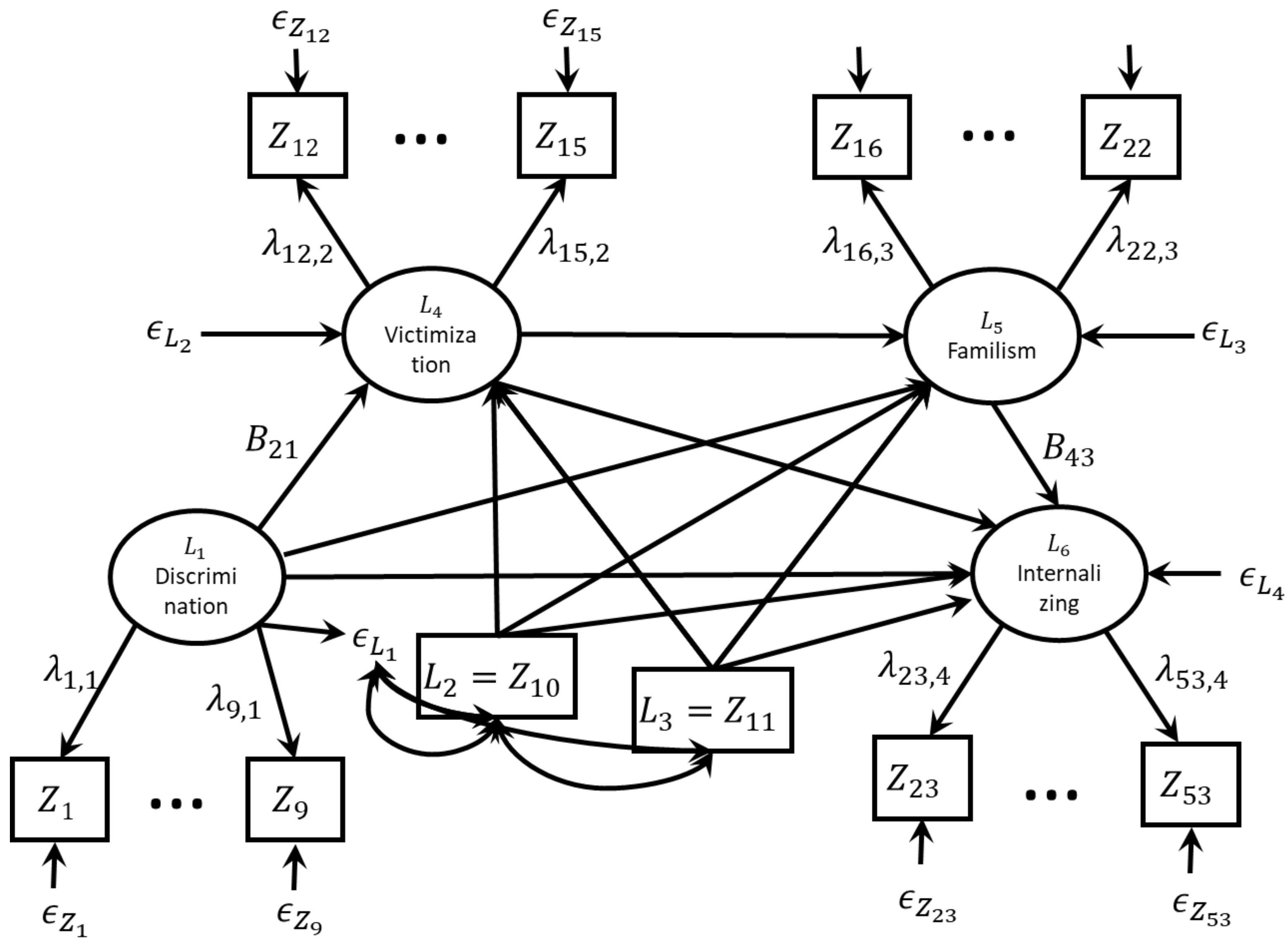
- N_θ -rule
- Three-indicator rule
- Two-indicator rule

Step 2. Treat the latent variable model as if it is an observed variable model, is it identified? If yes, model is identified.

Can apply all observed variable rules here:

- N_θ-rule
- Null-B_{YY} rule
- Recursive Rule
- Rank and order conditions

Applying the Two-Step Rule



Two-Indicator Rule

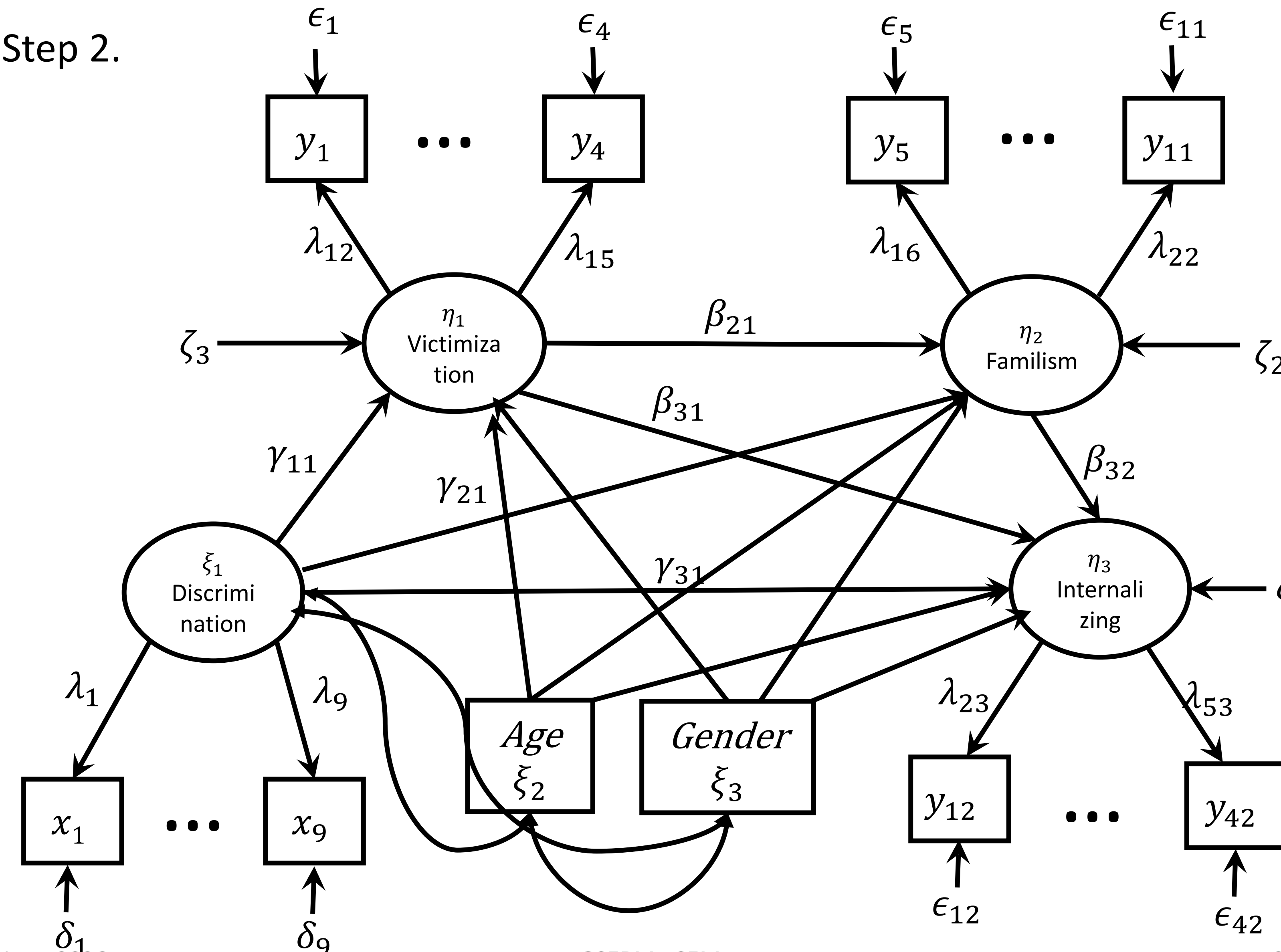
All latent factors have at least two indicators*

*Age and gender do not have two indicators, but have sufficient constraints to be identified.

Pass the two-indicator rule → Move to Step 2

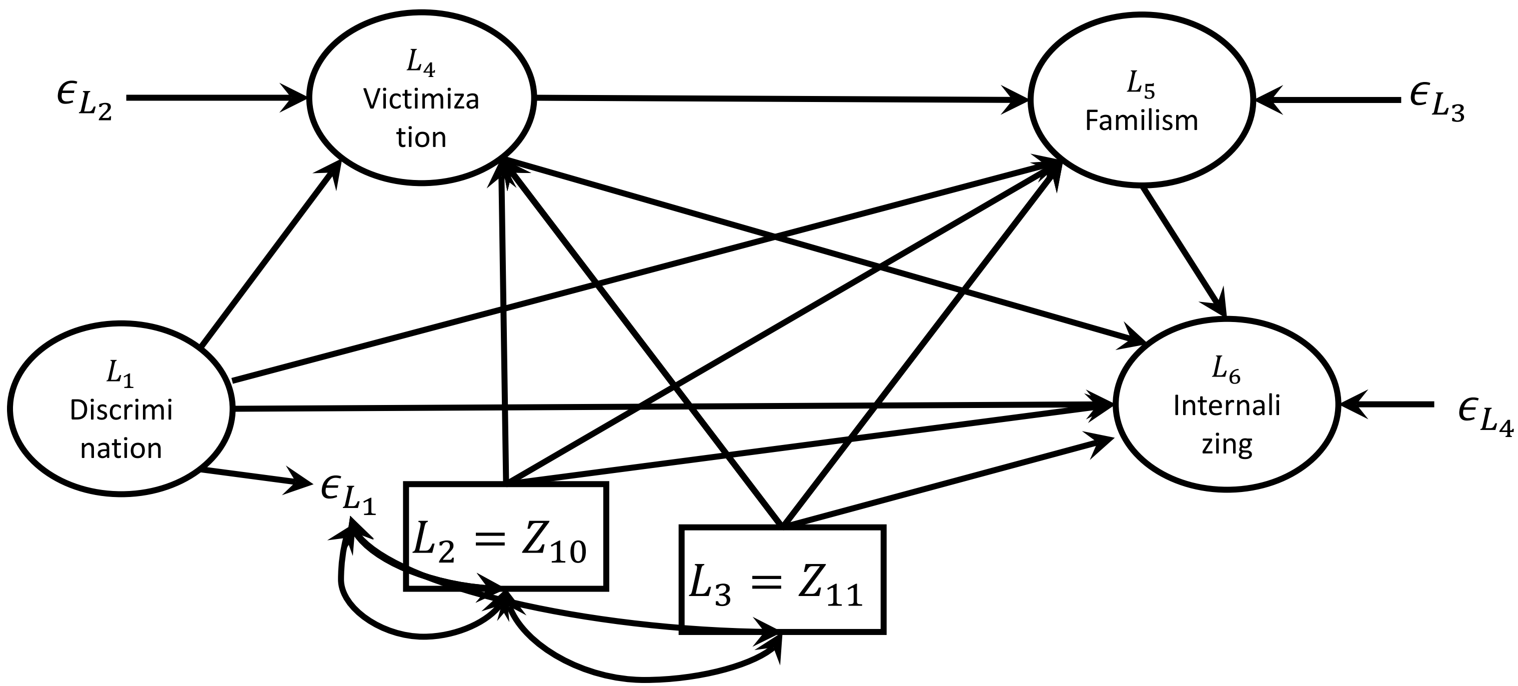
Applying the Two-Step Rule

Step 2.



Applying the Two-Step Rule

Step 2.



Is it Identified?

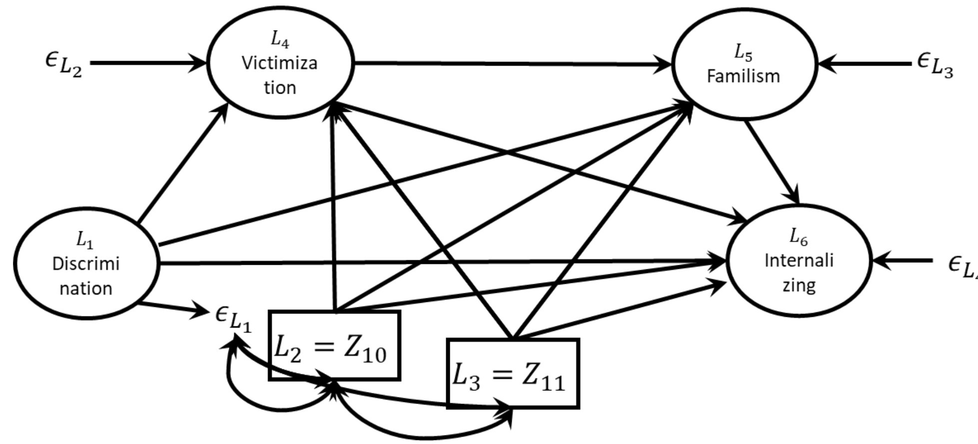
- N_θ-rule

  $$\frac{1}{2}(6)(6+1) = 21$$

  $$N_\theta = 21$$

  Passed!

- Null B_{YY}-Rule
  - B_{YY} is not null
- Recursive Rule
  - Model is not recursive
  - Model is identified!



Notice that the structural model is **just identified**, so no "misfit" can come from this part of the model

This means the whole model is identified!

Also there are no degrees of freedom in this part of the model

Estimation

Maximum Likelihood

Just as before we'll use maximum likelihood to estimate our model

$$F_{ML} = \log|\boldsymbol{\Sigma}(\boldsymbol{\theta})| + tr(\boldsymbol{S}\boldsymbol{\Sigma}(\boldsymbol{\theta})^{-1}) - \log|\boldsymbol{S}| - (N_Z)$$

These models tend to be more complex, so you may see that it takes more iterations to converge

One of the issues with "raw" maximum likelihood is that it uses *listwise deletion* for missing data

If any participant is missing on any variable, they are not used.

```
Estimator                                         ML
Optimization method                           NLMINB
Number of model parameters                       116

                                                Used       Total
Number of observations                           142         676
```

Full Information Maximum Likelihood

- Full Information Maximum Likelihood (FIML) estimation does not impute missing data, but instead uses all available information (i.e., all variables on which a subject provided data) to refine parameter estimates.
- By computing likelihood for all individuals and all missing data patterns, FIML adjusts parameter estimates by including more information than just the complete cases.
- Because log-likelihoods are independent for independent observations, can sum these likelihoods to get the overall likelihood for the sample.
- By fine-tuning parameter estimates in this way, the set of parameters that maximizes the overall likelihood is changed (including parameter estimates for variables with missing data).

$$LL = -\frac{Nk}{2}\log(2\pi) - \frac{N}{2}\log|\boldsymbol{\Sigma}(\theta)_i| - \frac{1}{2}\sum_{i=1}^{N}(\boldsymbol{Z}_i - \bar{\mathbf{Z}}_i)'\boldsymbol{\Sigma}(\boldsymbol{\theta})_i^{\;-1}(\boldsymbol{Z}_i - \bar{\mathbf{Z}}_i)$$

Evaluating the CFA Model

Why fit the CFA first?

- When fitting a complex SEM it may be adviseable to fit the appropriate CFA first
- This will help you evaluate possible mispecifications in the measurement model (before looking at the latent variable model)
- Can make modifications in the CFA model first
- Big differences in factor structure between CFA and SEM can indicate identification issues.

Fitting Model in lavaan

```
CFAmodelfit <- cfa(factormodel, data = dat)
summary(CFAmodel, fit.measures = TRUE)
```

```
Model Test User Model:

  Test statistic                               18959.064
  Degrees of freedom                                1268
  P-value (Chi-square)                             0.000

Model Test Baseline Model:

  Test statistic                             26735.184
  Degrees of freedom                              1326
  P-value                                        0.000

User Model versus Baseline Model:

  Comparative Fit Index (CFI)                    0.304
  Tucker-Lewis Index (TLI)                       0.272

Loglikelihood and Information Criteria:

  Loglikelihood user model (H0)             -31026.211
  Loglikelihood unrestricted model (H1)     -21546.679

  Akaike (AIC)                               62272.423
  Bayesian (BIC)                             62724.928
  Sample-size adjusted Bayesian (BIC)        62375.828

Root Mean Square Error of Approximation:

  RMSEA                                          0.176
  90 Percent confidence interval - lower         0.173
  90 Percent confidence interval - upper         0.178
  P-value RMSEA <= 0.05                          0.000

Standardized Root Mean Square Residual:

  SRMR                                           0.063
```

Model fits significantly worse than saturated model

CFI and TLI are low, suggesting poor fit

RMSEA is high, suggesting poor fit

SRMR is actually ok, but majority rules

Check the Results for Red Flags

Go over to results in R

```
Variances:
                   Estimate   Std.Err  z-value  P(>|z|)
   .safe1             1.115    0.076   14.746    0.000
   .safe2             0.520    0.039   13.298    0.000
   .safe8             0.426    0.032   13.401    0.000
   .safe10            0.411    0.033   12.500    0.000
   .safe13            0.277    0.022   12.638    0.000
   .safe15            0.597    0.046   13.046    0.000
   .safe16            0.143    0.011   13.612    0.000
   .safe20            0.107    0.008   13.697    0.000
   .safe21            0.301    0.021   14.189    0.000
   .bully4        34393.034 2992.053   11.495    0.000
   .bully5        34092.308 3103.547   10.985    0.000
   .bully6        34093.657 3103.682   10.985    0.000
   .bully7        34159.613 3109.834   10.984    0.000
   .FS1               0.390    0.029   13.520    0.000
   .FS2               0.663    0.047   14.026    0.000
   .FS3               0.353    0.029   12.337    0.000
   .FS4               0.397    0.033   12.058    0.000
   .FS5               0.281    0.024   11.803    0.000
   .FS6               0.345    0.026   13.443    0.000
   .FS7               0.674    0.050   13.356    0.000

    Disc                0.080    0.032    2.489    0.013
    Vict            46592.202 5244.502    8.884    0.000
    Famil               0.419    0.049    8.582    0.000
    Internal            0.167    0.023    7.213    0.000
```

This looks concerning

Notice that the bully items are the indicators for Vict

Try a Model without Bully

```
nobullymodel <- "
Disc =~ safe1 + safe2 + safe8 + safe10 + safe13 + safe15 + safe16 +
safe20 + safe21
Famil =~FS1+FS2+FS3+FS4+FS5+FS6+FS7
Internal =~
ysr_prob_14+ysr_prob_29+ysr_prob_30+ysr_prob_31+ysr_prob_32+ysr_pro
b_33+ysr_prob_35+ysr_prob_45+ysr_prob_50+ysr_prob_52+ysr_prob_71+ys
r_prob_91+ysr_prob_112+ysr_prob_5+ysr_prob_42+ysr_prob_65+ysr_prob_
69+ysr_prob_75+ysr_prob_102+ysr_prob_103+ysr_prob_111+ysr_prob_47+y
sr_prob_51+ysr_prob_54+ysr_prob_56a+ysr_prob_56b+ysr_prob_56c+ysr_p
rob_56d+ysr_prob_56e+ysr_prob_56f+ysr_prob_56g+ysr_prob_56h
"
nobullyfit <- cfa(nobullymodel, data = dat)
summary(nobullyfit, fit.measures = TRUE)
```

```
User Model versus Baseline Model:

  Comparative Fit Index (CFI)                    0.818
  Tucker-Lewis Index (TLI)                       0.809
```

```
Root Mean Square Error of Approximation:

  RMSEA                                          0.052
  90 Percent confidence interval - lower         0.050
  90 Percent confidence interval - upper         0.054
  P-value RMSEA <= 0.05                          0.056

Standardized Root Mean Square Residual:

  SRMR                                           0.061
```

This looks much better!

What's the deal with the bully items?

```
onlybullymodel <- "
Vict =~ bully4+bully5+bully6+bully7"
onlybullyfit <- cfa(onlybullymodel, data = dat)
summary(onlybullyfit, fit.measures = TRUE)
```

```
Model Test User Model:

  Test statistic                                 0.319
  Degrees of freedom                                 2
  P-value (Chi-square)                           0.852


User Model versus Baseline Model:

  Comparative Fit Index (CFI)                    1.000
  Tucker-Lewis Index (TLI)                       1.008

Root Mean Square Error of Approximation:

  RMSEA                                          0.000
  90 Percent confidence interval - lower         0.000
  90 Percent confidence interval - upper         0.075
  P-value RMSEA <= 0.05                          0.909

Standardized Root Mean Square Residual:

  SRMR                                           0.005
```

Everything actually looks pretty good?

```
Used       Total
 211         676
```

Only about a third of the sample took the bully scale! D-:

A decision point

There are some options that you might consider here:

- Analyze only the people with complete bully items
- Proceed anyway…?
- Simplify the model to exclude the bullying items

We’re going to proceed anyway, mostly so you can see what happens. In reality I might simplify.

Ultimately, we found that the bully items were zero inflated, so their distribution is very not normal, we can dichotomize them for better performance.

The Full Model

Evaluating Full Model in lavaan

```
SEMmodelfit <- sem(c(factormodel, latentvariables), data = dat, missing =
"FIML")
summary(SEMmodelfit, fit.measures=TRUE)
```

```
Model Test User Model:

  Test statistic                              3645.212
  Degrees of freedom                              1366
  P-value (Chi-square)                           0.000

Model Test Baseline Model:

  Test statistic                             13292.016
  Degrees of freedom                              1430
  P-value                                        0.000

User Model versus Baseline Model:

  Comparative Fit Index (CFI)                    0.808
  Tucker-Lewis Index (TLI)                       0.799

Loglikelihood and Information Criteria:

  Loglikelihood user model (H0)             -28459.644
  Loglikelihood unrestricted model (H1)     -26637.038

  Akaike (AIC)                               57255.288
  Bayesian (BIC)                             58014.008
  Sample-size adjusted Bayesian (BIC)        57480.592

Root Mean Square Error of Approximation:

  RMSEA                                          0.050
  90 Percent confidence interval - lower         0.048
  90 Percent confidence interval - upper         0.052
  P-value RMSEA <= 0.05                          0.603

Standardized Root Mean Square Residual:

  SRMR                                           0.063
```

Overall, not too shabby, could look for points of improvement

```
.bully4        0.178    0.025    7.009    0.000
.bully5        0.060    0.022    2.759    0.006
.bully6        0.298    0.034    8.741    0.000
.bully7        0.221    0.022   10.162    0.000
```

Scary variances actually went away

Research Questions

- Does a model where discrimination affects peer victimization, familism, and internalizing; peer victimization affects familism and internalizing; and familism affects internalizing **fit the data well**?

  Pretty well, but not great

- What is the **indirect effect** of discrimination on internalizing symptoms through peer victimization and familism in serial?
- Does a model which only allows for the serial indirect effect of discrimination on familism, and no other indirect (or direct) effects, fit better/well?

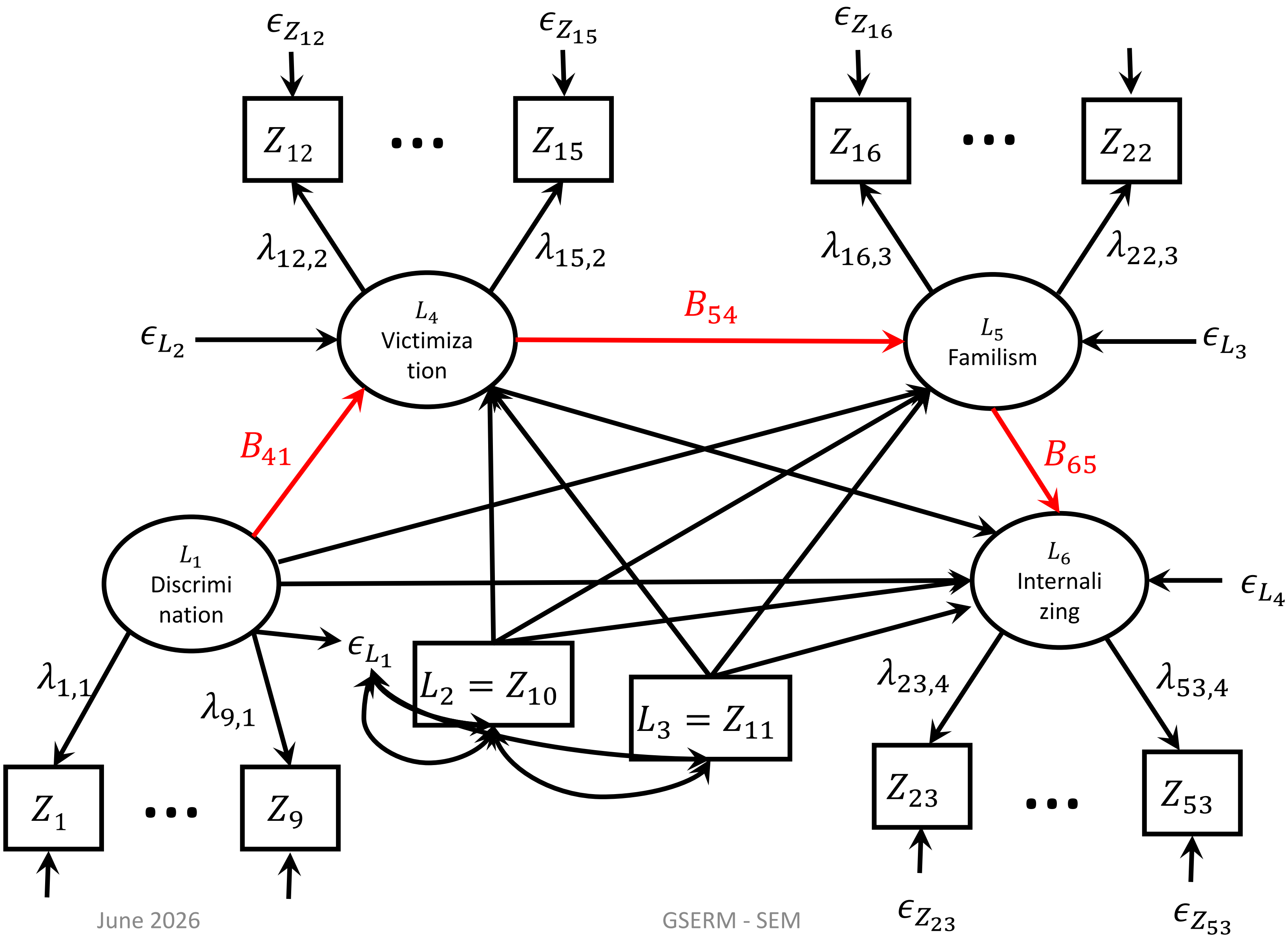
Indirect Effects

Indirect effects are quantifications of the effect of one variable on another through one or more variables.

Indirect effects are quantified as the path tracing (product of coefficients) from one variable to another.

Indirect effects assume the relationships among variables are **causal** (i.e., they don’t mean anything if the relationships are not causal), but do not provide evidence that these relationships are causal.

Indirect effect of discrimination on internalizing through both victimization and familism: $B_{41}\mathrm{B}_{54}\mathrm{B}_{65}$



Estimating Indirect Effects in Lavaan

Lavaan allows you to define new parameters of interest by combining existing parameters.

```
latentvariables2 <- "
Vict ~ B41*Disc+Age+Gender
Famil ~ Disc+B54*Vict+Age+Gender
Internal ~ Disc+Vict+B65*Famil+Age+Gender
"
```

Name the parameters that you need

```
indirectmodel <- "
serialind := B41*B54*B65
"
```

Combine them in the way you want

Re-estimating lavaan model

```
SEMmodelfit2 <- sem(c(factormodel, latentvariables2,
indirectmodel), data = dat, missing = "FIML")

summary(SEMmodelfit2)
```

```
Defined Parameters:
                   Estimate  Std.Err  z-value  P(>|z|)
    serialind         0.041    0.016    2.518    0.012
```

Lavaan automatrically provides a standard error, z-value and p-value

DO NOT USE THESE!

These values are estimated based on the “delta-method” which does not work well for products of coefficients in finite samples.

Instead, we will use bootstrap confidence intervals!

Bootstrapping in Lavaan

```
SEMmodelfit3 <- sem(c(factormodel, latentvariables2,
indirectmodel), se = "bootstrap", data = dat, bootstrap = 1000)

summary(SEMmodelfit3, standardized = TRUE)
```

Running this code will take a long time!

```
Defined Parameters:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
    serialind         0.041    0.023    1.769    0.077    0.031    0.031
```

Notice the standard error is a little different, this is a **bootstrap standard error**.

This test is still not valid, because it assumes the indirect effect has a normal distribution, and calculates the *Z* and *p* values based on this assumption

Standardized serial indirect effect is .03 (itty bitty)

Bootstrapping in Lavaan

```
paramsSEM3 <- parameterEstimates(SEMmodelfit3)

paramsSEM3[182,]
```

```
> paramsSEM3[182,]
          lhs op                        rhs      label   est    se     z pvalue ci.lower ci.upper
182 serialind := gamma11*beta21*beta32 serialind 0.041 0.023 1.769  0.077    0.005    0.091
```

Notice the point estimate doesn't change, and this is the bootstrap SE

95% CI does not include zero → significant indirect effect at $\alpha = .05$

```
parameterEstimates(SEMmodelfit3, level = .99)[182,]
```

You can change the level of the confidence interval with the level argument, here is a 99% CI

Research Questions

- Does a model where discrimination affects peer victimization, familism, and internalizing; peer victimization affects familism and internalizing; and familism affects internalizing **fit the data well**?

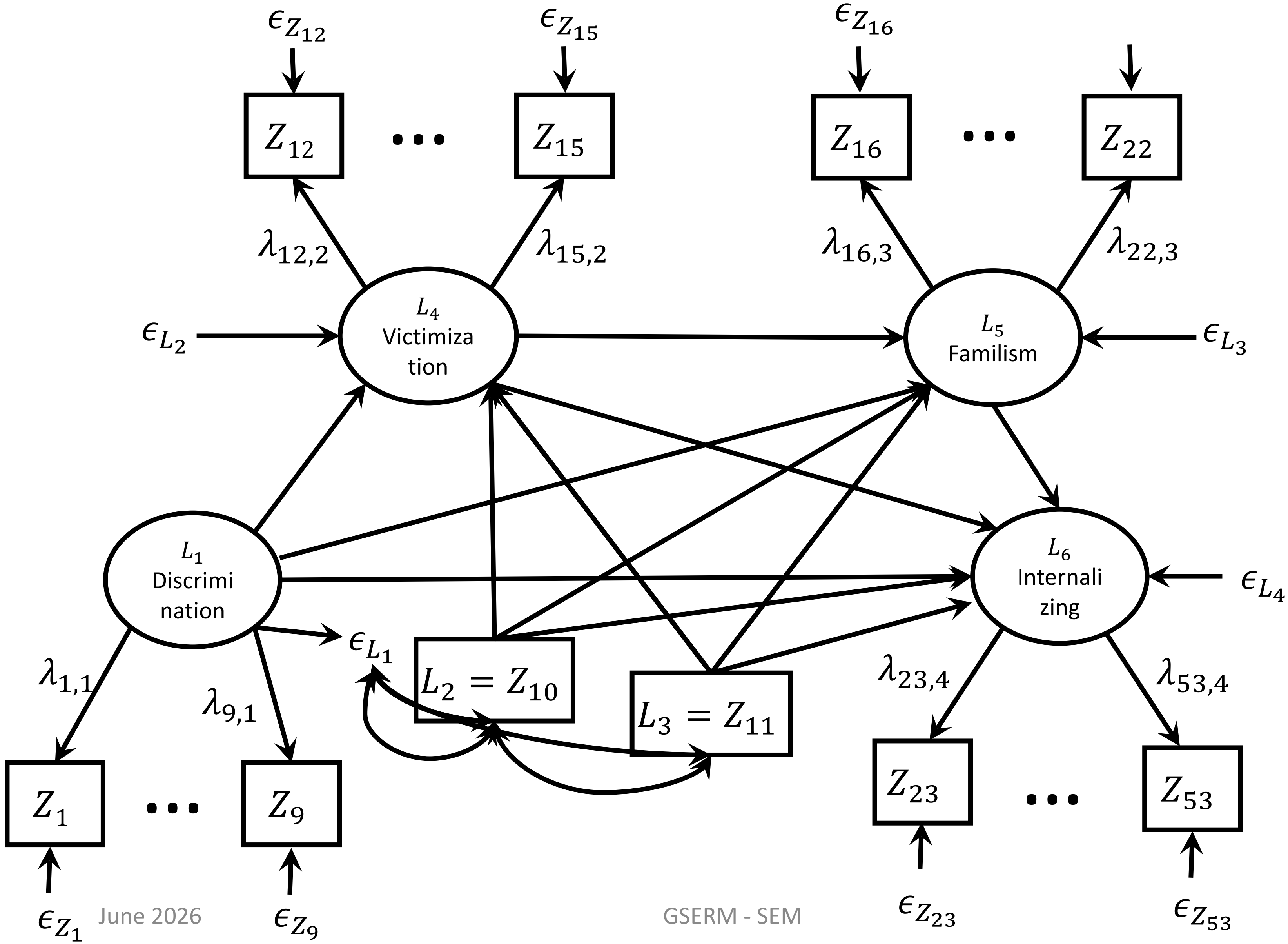
  Pretty well, but not great

- What is the **indirect effect** of discrimination on internalizing symptoms through peer victimization and familism in serial?

  Small, but significant!

- Does a model which only allows for the serial indirect effect of discrimination on familism, and no other indirect (or direct) effects, fit better/well?

The restricted model



Fitting the Restricted Model

```
# restricted model
restrictedlatentvariables <- "
Vict ~ Disc+Age+Gender
Famil ~ Vict+Age+Gender
Internal ~ Famil+Age+Gender
"

restrictedSEMmodelfit <- sem(c(factormodel,
restrictedlatentvariables, indirectmodel), data = dat, missing =
"FIML")
summary(restrictedSEMmodelfit, fit.measures = TRUE)
```

Removed all but the “prior” predictor in the model

Does the New Model Fit "Well"?

```
Model Test User Model:

  Test statistic                                3731.862
  Degrees of freedom                                1369
  P-value (Chi-square)                             0.000

Model Test Baseline Model:

  Test statistic                               13292.016
  Degrees of freedom                                1430
  P-value                                          0.000

User Model versus Baseline Model:

  Comparative Fit Index (CFI)                      0.801
  Tucker-Lewis Index (TLI)                         0.792

Loglikelihood and Information Criteria:

  Loglikelihood user model (H0)               -28502.969
  Loglikelihood unrestricted model (H1)       -26637.038

  Akaike (AIC)                                 57335.937
  Bayesian (BIC)                               58081.109
  Sample-size adjusted Bayesian (BIC)          57557.218

Root Mean Square Error of Approximation:

  RMSEA                                            0.051
  90 Percent confidence interval - lower           0.049
  90 Percent confidence interval - upper           0.052
  P-value RMSEA <= 0.05                            0.324

Standardized Root Mean Square Residual:

  SRMR                                             0.082
```

Fit looks fairly comparable to the other "less restrictive" model

Indirect Effect

```
Defined Parameters:
                   Estimate  Std.Err  z-value  P(>|z|)
    serialind         0.136    0.034    3.958    0.000
```

The serial indirect effect is much bigger!

What happened?

- Because there are omitted paths, their explained variance will get reassigned elsewhere in the model. This can **bias** indirect effect estimates.
- We cannot know if this estimate is biased or not.
- This is why it’s common to estimate a just identified latent variable model, for indirect effects.

Model Comparison

```
lavTestLRT(SEMmodelfit2, restrictedSEMmodelfit)
```

```
Chi-Squared Difference Test

                        Df   AIC   BIC  Chisq Chisq diff Df diff Pr(>Chisq)
SEMmodelfit2          1366 57255 58014 3645.2
restrictedSEMmodelfit 1369 57336 58081 3731.9      86.65       3  < 2.2e-16 ***
```

What is the conclusion?

The restricted model fits significantly worse than the less restricted model.

This suggests that perhaps some of those omitted paths are needed for appropriate representation of the data generating process

Research Questions

- Does a model where discrimination affects peer victimization, familism, and internalizing; peer victimization affects familism and internalizing; and familism affects internalizing **fit the data well**?

  Pretty well, but not great

- What is the **indirect effect** of discrimination on internalizing symptoms through peer victimization and familism in serial?

  Small, but significant!

- Does a model which only allows for the serial indirect effect of discrimination on familism, and no other indirect (or direct) effects, fit better/well?

  No, some of those indirect/direct effects are needed

Can we determine variable order based on model fit?

A common question which arises out of SEMs is whether we can determine causal variable order based on model fit.

Can we distinguish between:

- Discrimination → Peer Victimization → Familism → Internalizing
- Discrimination → Familism → Peer Victimization → Internalizing

These two models are not nested, so we cannot compare based on model fit.

Both models are **just identified**, so there is no differentiation there.

Typically, reordering the variables results in the same implied covariance matrix for the data. (Thoemmes, 2015)

Thank you!

- Thank you for your time this week!
- Please fill our your course evaluations
- I’ll post some slides on reporting recommendations in SEM
- Feel free to reach out about future questions